

Cost of Capital Assignment

Q1. The future earnings, dividends, and common stock price of Callahan Technologies Inc. are expected to grow 6% per year. Callahan's common stock currently sells for \$22.00 per share, its last dividend was \$2.00, and it will pay a \$2.12 dividend at the end of the current year.

- a. Using the DCF approach, what is its cost of common equity?

$$\begin{aligned} r_e &= \frac{D_1}{P_0} + g \\ &= \frac{\$2.12}{22} + 0.06 = 15.64\% \end{aligned}$$

- b. If the firm's beta is 1.2, the risk-free rate is 6%, and the average return on the market is 13%, what will be the firm's cost of common equity using the CAPM approach?

$$\begin{aligned} r_s &= r_f + (r_m - r_f) \beta_L \\ &= 6\% + (13\% - 6\%) 1.2 \\ &= 14.4\% \end{aligned}$$

- c. If the firm's bonds earn a return of 11%, based on the bond-yield-plus-risk-premium approach, what will be r_s ? Use the midpoint of the risk premium range discussed in Section 10-5 in your calculations.

$$\begin{aligned} r_s &= \text{Bond yield} + \text{Risk Premium} \\ &= 11\% + (13\% - 6\%) \\ &= 18\% \end{aligned}$$

- d. If you have equal confidence in the inputs used for the three approaches, what is your estimate of Callahan's cost of common equity?

$$r_s = \frac{15.64\% + 14.4\% + 18\%}{3} = 16.01\%$$

Q2. The Evanec Company's next expected dividend, D_1 , is \$3.18; its growth rate is 6%; and its common stock now sells for \$36.00. New stock (external equity) can be sold to net \$32.40 per share.

a. What is Evanec's cost of retained earnings, r_s ?

$$r_s = \frac{D_1}{P_0} + g = \frac{3.18}{36} + 0.06 = 14.83\%$$

b. What is Evanec's percentage flotation cost, F ?

$$\left(\frac{36 - 32.40}{32.40} \right) = 11.1\%$$

c. What is Evanec's cost of new common stock, r_e ?

$$r_s = \frac{D_1}{P_0(1-F)} + g = \frac{3.18}{32.40(1-0.111)} + 0.06 = 17.04\%$$

Q3. The Paulson Company's year-end balance sheet is shown below. Its cost of common equity is 14%, its before-tax cost of debt is 10%, and its marginal tax rate is 40%. Assume that the firm's long-term debt sells at par value. The firm's total debt, which is the sum of the company's short-term debt and long-term debt, equals \$1,167. The firm has 576 shares of common stock outstanding that sell for \$4.00 per share. Calculate Paulson's WACC using market-value weights.

$$MV \text{ of Equity} = 576 \times \$4 = \$2304$$

$$MV \text{ of Debt} = \$1167 \rightarrow \text{Total } 3471$$

$$w_d r_d (1-T) + w_c r_s$$

$$WACC = \frac{1167}{3471} 10\% (1-0.4) + \frac{2304}{3471} \times 0.14$$

$$= 0.0202 + 0.0929 = 11.32\%$$

Q4. Kahn Inc. has a target capital structure of 60% common equity and 40% debt to fund its \$10 billion in operating assets. Furthermore, Kahn Inc. has a WACC of 13%, a before-tax cost of debt of 10%, and a tax

rate of 40%. The company's retained earnings are adequate to provide the common equity portion of its capital budget. Its expected dividend next year (D_1) is \$3, and the current stock price is \$35.

- a. What is the company's expected growth rate?

$$WACC = w_d r_d (1-T) + w_e r_s$$

$$13\% = 0.4 \times 10\% (1-0.4) + 0.6 r_s$$

$$10.6\% = 0.6 r_s$$

$$r_s = 17.67\%$$

- b. If the firm's net income is expected to be \$1.1 billion, what portion of its net income is the firm expected to pay out as dividends?

$$g = (1 - \text{Div P0}) \times ROE$$

$$g = (1 - \text{Div P0}) \times \text{NI/Equity}$$

$$9.099\% = (1 - \text{Div P0}) \times \frac{1100}{6000}$$

$$9.099\% = (1 - \text{Div P0}) \times 0.1833$$

$$0.4964 = 1 - \text{Div P0}$$

$$\text{Div P0} = 50.36\%$$

$$r_s = \frac{D_1}{P_0} + g$$

$$17.67\% = \frac{3}{35} + g$$

$$g = 9.099\%$$

Ex. Madsen Motors's bonds have 23 years remaining to maturity. Interest is paid annually, they have a \$1,000 par value, the coupon interest rate is 9%, and the yield to maturity is 11%. What is the bond's current market price?

$$11\% = \frac{90 + \frac{1000 - PV}{23}}{\frac{1000 + PV}{2}}$$

$$PV = -\$834.67$$

Q7. Bond A is non-callable and has 25 years to maturity, 9% annual coupon and \$1000 par value. Your required rate of return on bond X is 10%, if you plan to hold it for 5 years. You expect that in 5 years, the interest rate on 20 years bond with the same risk is 8%. How much should you be willing to pay for Bond X today?

Bond A

$$N = 20$$

$$CP = 90$$

$$FV = 1000$$

$$PV = \text{Solve} = 617.58$$

$$I = 10\%$$

Bond X

$$N = 5 \text{ yrs}$$

$$I = 8\%$$

$$CP = 90$$

$$FV = 1000$$

$$PV = \text{Solve} = -321.24$$

Q8. Nesmith Corporation's outstanding bonds have a \$1,000 par value, a 8% semiannual coupon, 14 years to maturity, and an 11% YTM. What is the bond's price?

$$N = 28 \text{ yrs}$$

$$YTM = 11\% / 2 = 5.5\%$$

$$\text{Coupon PMT} = \frac{8\%}{2} = 4\%$$

$$FV = 1000$$

$$PV = \text{Value} = -341.54$$